POISSON DEFORMATION OF DEL PEZZO SURFACES

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Abstract. We study Poisson deformations of the simple elliptic surface singularities of types $\widetilde{E}_6$, $\widetilde{E}_7$, and $\widetilde{E}_8$. Their classical semiuniversal deformations are realized by families of affine del Pezzo surfaces. We explain how these families admit a natural Poisson enhancement: adjoining one parameter that rescales the Jacobian Poisson bracket gives a semiuniversal Poisson deformation. The proof is based on a computation of the Poisson cohomology of the corresponding singular hypersurface algebra, using a differential graded resolution and an associated spectral sequence.

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1. Introduction

A deformation describes how a geometric object changes as the coefficients of its defining equations vary. In this paper, the objects are affine surfaces with an isolated singular point. We study deformations that also keep track of a Poisson bracket, an operation on functions that is compatible with multiplication and satisfies the Jacobi identity.

The geometry comes from del Pezzo surfaces, a basic class of smooth projective algebraic surfaces. Choose such a surface S and a smooth anticanonical curve $E \subset S$. The curve E is elliptic, and removing it gives an affine surface

$$X := S \setminus E.$$

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A section of the anticanonical bundle whose zero divisor is E determines a Poisson bracket on S. On X, this bracket is nondegenerate and corresponds to a symplectic form. These geometric facts are explained in section 3.

For del Pezzo surfaces of type E_ℓ, where $\ell = 6, 7, 8$, the affine surface X admits an embedding into $\mathbb{C}^3$. Its defining equation can be written in the form

$$\phi^{\tau,\nu}_{P,Q,R}(x, y, z) = \tau xyz + P(x) + Q(y) + R(z) + \nu,$$

where

$$P(x) = \frac{x^p}{p} + \alpha_1 x^{p-1} + \cdots + \alpha_{p-1} x,$$

and similarly for $Q(y)$ and $R(z)$. The triples (p, q, r) are

$$(3, 3, 3), \qquad (4, 4, 2), \qquad (6, 3, 2)$$

in types E_6, E_7, E_8, respectively.

When the lower-order parameters vanish and τ is chosen so that the singularity is isolated, the equation

$$\phi^\tau(x, y, z) = \tau xyz + \frac{x^p}{p} + \frac{y^q}{q} + \frac{z^r}{r} = 0$$

defines a surface with an isolated simple elliptic singularity of type $\widetilde{E}_\ell$ at the origin. This singular surface is the central fiber of a family whose general fibers are smooth affine del Pezzo surfaces. A classical result states that the family obtained by varying the coefficients of P, Q, R, together with τ and ν, is a flat μ-parameter family of affine del Pezzo surfaces and gives a semiuniversal deformation of the corresponding simple elliptic singularity. Here

$$\mu = \dim_{\mathbb{C}} \mathbb{J}(\phi^\tau) = p + q + r - 1$$

is the Milnor number. Here $\mathbb{J}(\phi^\tau)$ is the Jacobi ring, the quotient of $\mathbb{C}[x, y, z]$ by the ideal generated by the three partial derivatives of ϕ^τ; see section 3.2. We recall the deformation theorem in theorem 3.8; see [EG10, §§2.4–2.5, especially Theorem 2.5.3(ii)], and [Loo84, Chapter 6] for the general deformation-theoretic framework.

Each hypersurface in this family also carries a natural Poisson structure. To construct it, start with a polynomial $\phi \in \mathbb{C}[x, y, z]$ and define the Jacobian Poisson bracket by

$$\{x, y\}_\phi = \frac{\partial \phi}{\partial z}, \qquad \{y, z\}_\phi = \frac{\partial \phi}{\partial x}, \qquad \{z, x\}_\phi = \frac{\partial \phi}{\partial y}.$$

The polynomial ϕ Poisson-commutes with every polynomial: $\{\phi, f\}_\phi = 0$ for all $f \in \mathbb{C}[x, y, z]$. Consequently, the quotient

$$\mathscr{B}_\phi := \mathbb{C}[x, y, z]/(\phi)$$

inherits a Poisson algebra structure. Thus, the classical deformation of the hypersurface $\phi^\tau = 0$ naturally gives a family of Poisson algebras.

The main result discussed in this paper is theorem 4.1. It states that the family

$$\left\{ \mathscr{B}_\phi \;\middle|\; \phi = c\, \phi^{\tau,\nu}_{P,Q,R}, \quad c \in \mathbb{C}^\times \right\}$$

is a semiuniversal Poisson deformation of $\mathscr{B}_{\phi^\tau}$. This is the Poisson enhancement of the classical result in theorem 3.8. After forgetting the Poisson bracket, one recovers the classical μ-parameter family of affine del Pezzo surfaces. The additional parameter c does not change the hypersurface $\{\phi = 0\}$, but it rescales the induced Poisson bracket. Consequently, the Poisson deformation space has $\mu + 1$ independent infinitesimal directions. This result is due to Etingof and Ginzburg; see [EG10, Theorem 2.5.3(iii) and Remark 2.5.5].

1.1. **Deformations and cohomology.** We briefly recall the deformation-theoretic ideas used in this paper; see also [EG10, §2.1]. We give an informal account of the main ideas and introduce the notion of a semiuniversal deformation used below.

Let A be an algebraic object, such as a commutative algebra or a Poisson algebra. Informally, a deformation of A is a flat family $\mathcal{A}$ over a pointed parameter space $(S, 0)$ whose fiber over 0 is A. Flatness expresses the condition that the members of the family vary continuously in an algebraic sense and that no unexpected relations appear or disappear under specialization.

The simplest deformations are first-order deformations, parametrized by the one-point scheme

$$\operatorname{Spec}\bigl(\mathbb{C}[\varepsilon]/(\varepsilon^2)\bigr).$$

Its coordinate ring is the algebra of dual numbers. For example, if the commutative multiplication is held fixed, a first-order deformation of a Poisson bracket has the form

$$\{f, g\}_\varepsilon = \{f, g\} + \varepsilon\,\beta(f, g), \qquad \varepsilon^2 = 0.$$

The Leibniz and Jacobi identities, considered modulo ε^2, impose linear conditions on β. These conditions say precisely that β is a cocycle in the appropriate Poisson deformation complex. Two infinitesimal deformations that differ by an automorphism that is the identity modulo ε determine cohomologous cocycles. For the full deformation problem, in which both multiplication and bracket may vary, equivalence classes of first-order Poisson deformations are described by the second Poisson cohomology group $PH^2(A)$.

More generally, the usual principle of deformation theory is that second cohomology parametrizes infinitesimal deformation directions, whereas third cohomology contains the obstructions to extending a first-order deformation to higher orders. In the Poisson setting, this role is played by $PH^2(A)$ and $PH^3(A)$, respectively.

Let $\mathcal{A} \to S$ be a deformation of an object $A = A_0$ over a pointed base $(S, 0)$. A tangent vector $v \in T_0S$ may be represented by a morphism

$$v\colon \operatorname{Spec}\bigl(\mathbb{C}[\varepsilon]/(\varepsilon^2)\bigr) \longrightarrow S$$

which maps the closed point to 0. Pulling back $\mathcal{A}$ along v gives a first-order deformation of A. Taking its deformation class defines the *Kodaira–Spencer map*

$$\mathrm{KS}_0\colon T_0S \longrightarrow H^2(A), \qquad v \longmapsto [v^*\mathcal{A}].$$

Here $H^2(A)$ denotes the second cohomology group governing the deformation problem under consideration. In particular, for deformations of a Poisson algebra A, the relevant group is its second Poisson cohomology group, so that we get the corresponding Kodaira–Spencer map

$$\mathrm{KS}_0\colon T_0S \longrightarrow PH^2(A).$$

Thus, the Kodaira–Spencer map measures the infinitesimal variation of the family. Its kernel consists of tangent directions along which the deformation is infinitesimally trivial, while its image consists of the first-order deformation classes realized by the given family.

Following the smooth-base convention of [EG10, Definition 2.1.2], the deformation $\mathcal{A} \to S$ is called *semiuniversal at* 0 if its Kodaira–Spencer map is an isomorphism:

$$\mathrm{KS}_0\colon T_0S \xrightarrow{\sim} H^2(A),$$

or, in the Poisson case,

$$\mathrm{KS}_0\colon T_0S \xrightarrow{\sim} PH^2(A).$$

Surjectivity means that every first-order deformation of A occurs in the family, whereas injectivity means that the base has no redundant first-order parameters. The prefix "semi" reflects the fact that a morphism from another deformation to the semiuniversal family need not be unique. Therefore, a

semiuniversal deformation may be viewed intuitively as a smallest family containing all infinitesimal deformation directions of A.

1.2. **Organization of the paper.** In section 2, we review polyvector fields, contraction operators, generalized Lie derivatives, the Schouten bracket, and Poisson bivectors. These identities provide the algebraic framework for the Poisson cohomology computations used later.

In section 3, we recall the geometry of del Pezzo surfaces, quasi-homogeneous surface singularities, and simple elliptic singularities. We also explain how affine del Pezzo surfaces arise as deformations of the corresponding elliptic cones and state the classical semiuniversal deformation theorem.

Finally, in section 4, we study the Poisson cohomology of the singular hypersurface algebra $\mathscr{B}_\phi$. Using a differential graded resolution, we construct a spectral sequence computing this cohomology. Together with Pichereau's calculation of the relevant Lichnerowicz–Koszul–Brylinski cohomology, this yields the description of $PH^2(\mathscr{B}_{\phi^\tau})$ and proves the main theorem.

2. Polyvector fields, Schouten brackets and Poisson structures

In this section, we introduce polyvector fields over an algebra and the Schouten bracket. We prove that a bivector defines a Poisson structure precisely when its Schouten bracket with itself vanishes; see proposition 2.8. This criterion explains the Poisson structures on del Pezzo surfaces discussed in section 3. Polyvector fields will also be used in section 4 to compute Poisson cohomology.

Throughout this section, A is a commutative unital $\mathbb{C}$-algebra, and we do not assume that $\operatorname{Spec} A$ is smooth. Let Ω^1_A be the module of Kähler differentials and put

$$\Omega^q_A := \bigwedge_A^q \Omega^1_A, \qquad \Omega^\bullet_A := \bigoplus_{q\geq 0} \Omega^q_A.$$

The universal derivation $d\colon A \to \Omega^1_A$ extends uniquely to the de Rham differential $d\colon \Omega^q_A \to \Omega^{q+1}_A$.

2.1. Polyvector fields and contraction.

Definition 2.1 (Polyvector fields)**.** For $p \geq 1$, let

$$\operatorname{AltDer}^p_{\mathbb{C}}(A, A)$$

denote the space of alternating $\mathbb{C}$-multilinear maps $A^p \to A$ that are derivations in each argument. We define

$$\mathfrak{X}^p(A) := \operatorname{AltDer}^p_{\mathbb{C}}(A, A).$$

We also set

$$\mathfrak{X}^0(A) := A, \qquad \mathfrak{X}^\bullet(A) := \bigoplus_{q\geq 0} \mathfrak{X}^q(A).$$

Indeed we have $\mathfrak{X}^p(A) \cong \operatorname{Hom}_A(\Omega^p_A, A)$. The isomorphism sends $X \in \operatorname{Hom}_A(\Omega^p_A, A)$ to the multi-derivation

$$(a_1, \ldots, a_p) \longmapsto X(da_1 \wedge \cdots \wedge da_p).$$

Conversely, a derivation in each argument factors through the universal derivation $d\colon A \to \Omega^1_A$ in each slot. Alternation then makes the resulting A-multilinear map factor uniquely through Ω^p_A. This constructs the inverse and explains why the isomorphism does not require smoothness. An element of $\mathfrak{X}^p(A)$ is called a p-polyvector field.

Remark 2.2. This definition is valid even when A is singular. In particular, we do not identify $\mathfrak{X}^p(A)$ with $\bigwedge_A^p \operatorname{Der}_{\mathbb{C}}(A) \cong \bigwedge_A^p \operatorname{Hom}_A(\Omega_A^1, A)$. When $\operatorname{Spec} A$ is smooth, these graded algebras are naturally isomorphic. We use definition 2.1 throughout.

One point to keep in mind is that, with definition 2.1, there does not *a priori* exist a wedge product $\wedge : \mathfrak{X}^p A \times \mathfrak{X}^q A \to \mathfrak{X}^{p+q} A$, unlike the case where we have a natural wedge product $\bigwedge_A^p \operatorname{Hom}_A(\Omega_A^1, A) \times \bigwedge_A^q \operatorname{Hom}_A(\Omega_A^1, A) \to \bigwedge_A^{p+q} \operatorname{Hom}_A(\Omega_A^1, A)$.

This motivates the contraction operators introduced below. Roughly speaking, the wedge product can be recovered from contraction by duality. We do not use this construction later, so we do not discuss it in detail.

We now define contraction without choosing a basis and without writing a polyvector as a sum of decomposable polyvectors. For $q \geq p$, let

$$\Delta_{p,q-p} \colon \Omega_A^q \longrightarrow \Omega_A^p \otimes_A \Omega_A^{q-p}$$

be the exterior coalgebra map given by

$$\Delta_{p,q-p}(\alpha_1 \wedge \cdots \wedge \alpha_q) = \sum_{\sigma \in \operatorname{Sh}(p,q-p)} \operatorname{sgn}(\sigma)(\alpha_{\sigma(1)} \wedge \cdots \wedge \alpha_{\sigma(p)}) \otimes (\alpha_{\sigma(p+1)} \wedge \cdots \wedge \alpha_{\sigma(q)}),$$

where $\operatorname{Sh}(p, q-p)$ denotes the set of $(p, q-p)$-unshuffles, that is, permutations $\sigma \in S_q$ such that $\sigma(1) < \sigma(2) < \cdots < \sigma(p)$ and $\sigma(p+1) < \sigma(p+2) < \cdots < \sigma(q)$.

Definition 2.3 (Contraction by a polyvector)**.** For $X \in \mathfrak{X}^p(A)$, define the contraction operator

$$i_X \colon \Omega_A^q \longrightarrow \Omega_A^{q-p}$$

to be zero when $q < p$, and, when $q \geq p$, to be

$$i_X := (X \otimes \mathrm{id}) \circ \Delta_{p,q-p}.$$

Equivalently,

$$i_X(a_0\, da_1 \wedge \cdots \wedge da_q) = a_0 \sum_{\sigma \in \operatorname{Sh}(p,q-p)} \operatorname{sgn}(\sigma) X(da_{\sigma(1)} \wedge \cdots \wedge da_{\sigma(p)}) \cdot da_{\sigma(p+1)} \wedge \cdots \wedge da_{\sigma(q)}.$$

Thus i_X is an A-linear operator of degree $-p$ on $\Omega_A^\bullet$.

The map $X \mapsto i_X$ is injective: its restriction to Ω_A^p is precisely the functional $X \colon \Omega_A^p \to A$.

There is a second, equally basis-free contraction, with the roles of forms and polyvectors reversed. If $\alpha \in \Omega_A^r$ and $X \in \mathfrak{X}^p(A)$ with $r \leq p$, define

$$(i_\alpha X)(\beta) := X(\alpha \wedge \beta), \qquad \beta \in \Omega_A^{p-r}. \tag{1}$$

Then $i_\alpha X \in \mathfrak{X}^{p-r}(A)$. In particular, if $\alpha = da_1 \wedge \cdots \wedge da_r$, then

$$i_\alpha X = i_{da_r} \cdots i_{da_1} X.$$

The type of the argument will always distinguish the two uses of i.

2.2. **The generalized Lie derivative and the Schouten bracket.** If D and E are homogeneous operators on $\Omega_A^\bullet$, of operator degrees $|D|$ and $|E|$, respectively, their graded commutator is

$$[D, E]_{\mathrm{gr}} := D \circ E - (-1)^{|D||E|} E \circ D.$$

Definition 2.4 (Generalized Lie derivative)**.** For $X \in \mathfrak{X}^p(A)$, define

$$L_X := [i_X, d]_{\mathrm{gr}} = i_X d - (-1)^p d i_X.$$

It is an operator of degree $1-p$ on $\Omega_A^\bullet$.

We now show that the operator identity suggested by Cartan calculus really does define a bracket on polyvector fields for an arbitrary A.

For $X \in \mathfrak{X}^p(A)$ and $Y \in \mathfrak{X}^q(A)$, with $p \geq 1$, set

$$(X \circ Y)(a_1, \ldots, a_{p+q-1}) := \sum_{\sigma \in \mathrm{Sh}(q, p-1)} \mathrm{sgn}(\sigma)\, X\big(Y(a_{\sigma(1)}, \ldots, a_{\sigma(q)}), a_{\sigma(q+1)}, \ldots, a_{\sigma(p+q-1)}\big). \tag{2}$$

When $q = 0$, the expression Y means the element $Y \in A$; when $p = 0$, we set $X \circ Y = 0$.

Proposition 2.5 (Operator characterization of the Schouten bracket)**.** *For $X \in \mathfrak{X}^p(A)$ and $Y \in \mathfrak{X}^q(A)$ with $p + q \geq 1$, the alternating map*

$$[X, Y]_{\mathrm{S}} := X \circ Y - (-1)^{(p-1)(q-1)} Y \circ X \tag{3}$$

is a multiderivation, hence an element of $\mathfrak{X}^{p+q-1}(A)$, and it satisfies

$$\boxed{i_{[X,Y]_{\mathrm{S}}} = [L_X, i_Y]_{\mathrm{gr}}.} \tag{4}$$

For $p = q = 0$, set $[X, Y]_{\mathrm{S}} = 0$. Equation (4) uniquely determines $[X, Y]_{\mathrm{S}}$.

Proof. The two Leibniz expansions in (3) have the same mixed terms with opposite signs; consequently their difference is a derivation in each argument. Alternation is built into the unshuffle sums. Thus (3) defines a member of $\mathfrak{X}^{p+q-1}(A)$.

To prove (4), apply both sides to a generator $a_0\, da_1 \wedge \cdots \wedge da_m$ of Ω_A^m. If $m \geq p+q-1$, expanding d and the two shuffle contractions gives

$$\begin{aligned}[L_X, i_Y]_{\mathrm{gr}}(a_0\, da_1 \wedge \cdots \wedge da_m) = a_0 \sum_{\sigma \in \mathrm{Sh}(p+q-1, m-p-q+1)} \mathrm{sgn}(\sigma) \\ \cdot [X, Y]_{\mathrm{S}}(a_{\sigma(1)}, \ldots, a_{\sigma(p+q-1)}) \\ \cdot da_{\sigma(p+q)} \wedge \cdots \wedge da_{\sigma(m)}.\end{aligned}$$

The terms in which d acts on a_0 cancel. The remaining terms are exactly $i_{[X,Y]_{\mathrm{S}}}$ by the definition of contraction. If $m < p + q - 1$, both sides vanish. Finally, uniqueness follows from the injectivity of $Z \mapsto i_Z$. □

In fact, we prefer to take (4) as the conceptual definition of the Schouten bracket, the unique polyvector whose contraction equals the graded commutator of L_X and i_Y. Formula (3) proves its existence without choosing a basis or assuming smoothness.

Proposition 2.6 (Cartan identities)**.** *For homogeneous polyvectors X, Y, one has*

$$[L_X, L_Y]_{\mathrm{gr}} = L_{[X,Y]_{\mathrm{S}}}, \tag{5}$$

$$[X, Y]_{\mathrm{S}} = -(-1)^{(p-1)(q-1)} [Y, X]_{\mathrm{S}}, \tag{6}$$

where p and q are the polyvector degrees of X and Y.

Proof. The graded Jacobi identity for graded commutators and $d^2 = 0$ give $[L_X, d]_{\mathrm{gr}} = 0$. Therefore

$$\begin{aligned}[L_X, L_Y]_{\mathrm{gr}} &= [L_X, [i_Y, d]_{\mathrm{gr}}]_{\mathrm{gr}} \\ &= [[L_X, i_Y]_{\mathrm{gr}}, d]_{\mathrm{gr}} = [i_{[X,Y]_{\mathrm{S}}}, d]_{\mathrm{gr}} = L_{[X,Y]_{\mathrm{S}}}.\end{aligned}$$

The shifted skew-symmetry is immediate from (3). □

The adjoint action on polyvectors is often written with the same Lie derivative notation,

$$L_X Y := [X, Y]_{\mathrm{S}}.$$

Equation (5) expresses the compatibility of this action with the operator L_X on differential forms.

For later use, notice that if $f \in A$ and $X \in \mathfrak{X}^p(A)$, then

$$[X, f]_{\mathrm{S}} = i_{df} X. \tag{7}$$

In particular, for a vector field $X \in \mathfrak{X}^1(A)$, $[X, f]_{\mathrm{S}} = X(f)$.

Corollary 2.7 (Mixed Cartan identity)**.** *Let $\pi \in \mathfrak{X}^2(A)$ and $f \in A$. With the convention*

$$L_P(Q) := [Q, P]_{\mathrm{S}},$$

one has

$$[L_\pi, i_{df}]_{\mathrm{gr}} = L_{i_{df}\pi}.$$

Equivalently,

$$L_\pi i_{df} + i_{df} L_\pi = L_{i_{df}\pi}.$$

Proof. For $Q \in \mathfrak{X}^q(A)$, the graded Jacobi identity and (6) give

$$\begin{aligned}\big(L_\pi i_{df} + i_{df} L_\pi\big)(Q) &= [[Q, f]_{\mathrm{S}}, \pi]_{\mathrm{S}} + [[Q, \pi]_{\mathrm{S}}, f]_{\mathrm{S}} \\ &= [Q, [f, \pi]_{\mathrm{S}}]_{\mathrm{S}}.\end{aligned}$$

Since $\deg f = 0$ and $\deg \pi = 2$, one has

$$[f, \pi]_{\mathrm{S}} = [\pi, f]_{\mathrm{S}}.$$

Moreover, by (7),

$$[\pi, f]_{\mathrm{S}} = i_{df}\pi.$$

Consequently,

$$\big(L_\pi i_{df} + i_{df} L_\pi\big)(Q) = [Q, i_{df}\pi]_{\mathrm{S}} = L_{i_{df}\pi}(Q).$$

□

2.3. **Poisson bivectors.** Let $\pi \in \mathfrak{X}^2(A)$. Define

$$\{f, g\}_\pi := \pi(df \wedge dg), \qquad f, g \in A. \tag{8}$$

Because π is an alternating biderivation, this operation is skew-symmetric and satisfies the Leibniz rule in each argument. Its Hamiltonian derivation is

$$X_f := \{f, -\}_\pi = [\pi, f]_{\mathrm{S}} = i_{df}\pi. \tag{9}$$

Proposition 2.8 (Poisson criterion)**.** *The bracket (8) is a Poisson bracket if and only if*

$$[\pi, \pi]_{\mathrm{S}} = 0.$$

More precisely, its Jacobiator is

$$\{f, \{g, h\}_\pi\}_\pi + \{g, \{h, f\}_\pi\}_\pi + \{h, \{f, g\}_\pi\}_\pi = -\frac{1}{2}[\pi, \pi]_{\mathrm{S}}(df \wedge dg \wedge dh). \tag{10}$$

Proof. Since contraction is injective, the three-vector $[\pi,\pi]_{\mathrm{S}}$ vanishes precisely when its contraction vanishes. It suffices to evaluate it on forms $df\wedge dg\wedge dh$, which generate Ω^3_A as an A-module. On such a three-form, contraction by a three-vector is simply evaluation. We therefore compute this evaluation and compare it with the Jacobiator.

Now by (4),

$$\begin{aligned} i_{[\pi,\pi]_S}(df\wedge dg\wedge dh) &= [L_\pi, i_\pi]_{\mathrm{gr}}(df\wedge dg\wedge dh)\\ &= ((i_\pi d - d i_\pi)i_\pi - i_\pi(i_\pi d - d i_\pi))(df\wedge dg\wedge dh). \end{aligned}$$

Since $d(df\wedge dg\wedge dh)=0$ and i_π^2 lowers the form degree by four, we have $i_\pi^2(df\wedge dg\wedge dh)=0$. The expression above therefore reduces to

$$2 i_\pi d i_\pi(df\wedge dg\wedge dh).$$

Expanding the contraction and comparing the resulting terms with the Jacobiator proves (10) and hence the proposition. □

3. Del Pezzo surfaces and isolated elliptic singularities

3.1. Del Pezzo surfaces.

Definition 3.1. A *del Pezzo surface* is a smooth projective surface S that is isomorphic either to the blow-up of $\mathbb{P}^2$ at ℓ sufficiently general points, where $0\le\ell\le 8$, or to $\mathbb{P}^1\times\mathbb{P}^1$.

Let K_S denote the canonical bundle of S, and let K_S^{-1} be the anticanonical bundle (the dual of K_S); that is,

$$K_S = \Lambda^2\Omega_S, \qquad K_S^{-1} = \Lambda^2 T_S.$$

If we take

$$A = \Gamma(S, T_S)$$

and apply the discussion in Section 2, then a regular section

$$\pi\in\Gamma(S, K_S^{-1}) = \mathfrak{X}^2 A$$

gives rise to a Poisson structure on A, since $[\pi,\pi]$ automatically vanishes in $\mathfrak{X}^3 A = 0$.

Definition 3.2. A regular section $\pi\in\Gamma(S,K_S^{-1})$ is said to be *non-degenerate* if its divisor of zeros is a reduced smooth curve.

Proposition 3.3. *When π is a non-degenerate section, the zero locus of π is an elliptic curve $E\subset S$, and $X := S\setminus E$ is affine.*

Proof. The first statement is a simple application of the adjunction formula

$$K_E \cong K_S\otimes\mathcal{O}_S(E)\big|_E.$$

By definition of E, $\mathcal{O}_S(-E)\cong\mathcal{I}_E$, the ideal sheaf of E, and we consider the map $\mathcal{O}_S\xrightarrow{\pi} K_S^{-1}$ given by the regular section π. After dualizing, this gives $K_S\xrightarrow{\cdot\pi}\mathcal{O}_S$, whose image is precisely the zero-locus ideal $\mathcal{I}_E$. Thus $K_S\cong\mathcal{I}_E\cong\mathcal{O}_S(-E)$, so $K_E\cong\mathcal{O}_E$. Hence the genus g of the reduced smooth curve E satisfies $2g-2=0$, namely $g=1$, and E is an elliptic curve.

The second statement follows from the two theorems quoted below. The first one is a well-known result from classical birational geometry. A proof using the birational classification of rational surfaces can be found in [Man86, Chapter IV, §24, Theorems 24.3–24.4]. The proof of the second theorem will be given below. □

Theorem 3.4. *A smooth projective surface S is del Pezzo if and only if its anticanonical bundle is ample.*

Theorem 3.5. *Suppose V is a projective variety and D is an effective ample divisor on V. Then $V \setminus Supp(D)$ is an affine variety.*

Proof. Since D is ample, there exists an integer $m > 0$ such that $\mathcal{O}_V(mD)$ is very ample. Let

$$\iota : V \hookrightarrow \mathbb{P}^N$$

be the closed embedding defined by the complete linear system $|mD|$. If s_D denotes the canonical section of $\mathcal{O}_V(D)$ whose zero divisor is D, then $s_D^m \in H^0(V, \mathcal{O}_V(mD))$ has zero divisor mD. Hence there is a hyperplane $H \subset \mathbb{P}^N$ such that

$$\iota^{-1}(H) = mD$$

scheme-theoretically. Since D and mD have the same support, we obtain

$$V \setminus \mathrm{Supp}(D) = V \setminus \mathrm{Supp}(mD) = \iota(V) \cap (\mathbb{P}^N \setminus H).$$

Now $\mathbb{P}^N \setminus H \cong \mathbb{A}^N$, and $\iota(V) \cap (\mathbb{P}^N \setminus H)$ is closed in $\mathbb{P}^N \setminus H$. It is therefore affine. □

We now consider the affine del Pezzo surface X and its coordinate ring $\mathbb{C}[X]$. The surface X has an algebraic symplectic form

$$\pi^{-1} \in \Gamma(S \setminus E, K_S),$$

which makes $\mathbb{C}[X]$ into a Poisson algebra; this is exactly the Poisson structure given by π after restricting to $X \subset S$. We will focus on this algebra in the sequel.

We denote the del Pezzo surface obtained by blowing up ℓ points in sufficiently general position in $\mathbb{P}^2$ by S_ℓ. We focus on the cases $\ell = 6, 7, 8$. In these cases a special feature is that the affine del Pezzo surfaces

$$X_\ell := S_\ell \setminus E$$

can be embedded into $\mathbb{C}^3$, as summarized by the following theorem.

Theorem 3.6. *The following statements hold.*

(i) The affine del Pezzo surface X_ℓ can be embedded into $\mathbb{C}^3$ when $\ell = 6, 7, 8$.

(ii) Its defining equation may be written in the form

$$\phi^{\tau,\nu}_{P,Q,R} := \tau xyz + P(x) + Q(y) + R(z) + \nu \in \mathbb{C}[x, y, z],$$

where

$$P(x) = \frac{1}{p}x^p + \alpha_1 x^{p-1} + \cdots + \alpha_{p-1}x,$$
$$Q(y) = \frac{1}{q}y^q + \beta_1 y^{q-1} + \cdots + \beta_{q-1}y,$$
$$R(z) = \frac{1}{r}z^r + \gamma_1 z^{r-1} + \cdots + \gamma_{r-1}z,$$

and $\tau \in \mathbb{C}^\times$, $\nu \in \mathbb{C}$. The triples (p, q, r) are

$$(3, 3, 3), \qquad (4, 4, 2), \qquad (6, 3, 2)$$

in the cases $\ell = 6, 7, 8$, respectively.

We refer to [EG10, Theorem 2.5.3(i)] for this description. The relation with elliptic cones is summarized in theorem 3.7.

Theorem 3.7. *The following statements describe the cone construction; see [EG10, §1.1 and the discussion preceding Theorem 2.5.3].*

(i) For a projective elliptic curve E, its affine cone C_E has an isolated elliptic singularity at the cone vertex.

(ii) For an affine del Pezzo surface $X = S \setminus E$, where S is the projective del Pezzo surface, X is a flat deformation of C_E.

This relates the description of affine del Pezzo surfaces to two questions: which elliptic cones occur, and how their singularities deform. The first question is addressed in proposition 3.11. For the second, a semiuniversal deformation of an isolated hypersurface singularity can be constructed by varying its defining polynomial. In our setting, the result is the following theorem.

Theorem 3.8. *The equations*

$$\phi^{\tau,\nu}_{P,Q,R}(x,y,z) = 0$$

give a flat μ-parameter family of affine del Pezzo surfaces of the corresponding type E_ℓ, providing a semiuniversal deformation of the isolated elliptic singularity of the corresponding type $\widetilde{E}_\ell$, where

$$\mu = p + q + r - 1, \qquad \ell = 6, 7, 8.$$

For a proof of this theorem, we refer to the standard construction of a semiuniversal deformation of an isolated hypersurface singularity from a basis of its Tjurina algebra; see [Loo84, Chapter 6, especially §6.C]. Since ϕ^τ is quasi-homogeneous, its Tjurina algebra agrees with its Jacobi ring. The relation with affine del Pezzo surfaces is explained in [EG10, Theorem 2.5.3(ii)]. We omit the details here.

3.2. **Quasi-homogeneous surface singularities.** We consider the coordinate ring $\mathbb{C}[x,y,z]$ of $\mathbb{C}^3$ as a nonnegatively graded algebra with homogeneous generators of degrees $\deg x = a$, $\deg y = b$, and $\deg z = c$, where a, b, c are positive integers.

Definition 3.9. A polynomial $\phi \in \mathbb{C}[x,y,z]$ is a weighted homogeneous (or quasi-homogeneous) polynomial of weighted degree $\deg \phi = d$ if and only if one has $\mathrm{eu}(\phi) = d \cdot \phi$, where

$$\mathrm{eu} := ax\frac{\partial}{\partial x} + by\frac{\partial}{\partial y} + cz\frac{\partial}{\partial z}, \tag{11}$$

denotes the *Euler vector field*.

For any polynomial $\phi \in \mathbb{C}[x,y,z]$ with an isolated singularity, its *Jacobi ring* is

$$\mathbb{J}(\phi) := \mathbb{C}[x,y,z] \Big/ \left(\frac{\partial\phi}{\partial x}, \frac{\partial\phi}{\partial y}, \frac{\partial\phi}{\partial z}\right).$$

If ϕ is weighted homogeneous of degree d and has an isolated critical point at the origin, then the origin is its only critical point. Furthermore, the Jacobi ring acquires a natural grading $\mathbb{J}(\phi) = \bigoplus_{m\geq 0} \mathbb{J}^{(m)}(\phi)$.

Proposition 3.10. *The Hilbert–Poincaré polynomial of this ring is*

$$\sum_{m\geq 0} u^m \cdot \dim \mathbb{J}^{(m)}(\phi) = \frac{(u^{d-a}-1)(u^{d-b}-1)(u^{d-c}-1)}{(u^a-1)(u^b-1)(u^c-1)}. \tag{12}$$

Proof. Put $S = \mathbb{C}[x,y,z]$, graded by $\deg x = a$, $\deg y = b$, and $\deg z = c$. Its Hilbert series is

$$H_S(u) = \frac{1}{(1-u^a)(1-u^b)(1-u^c)}.$$

Since ϕ has weighted degree d, its partial derivatives have degrees $d-a$, $d-b$, and $d-c$. The singularity is isolated, so the Jacobian ideal

$$I = (\phi_x, \phi_y, \phi_z)$$

has finite codimension in S. Hence its three homogeneous generators form a regular sequence. Quotienting a graded ring by a homogeneous non-zero-divisor of degree e multiplies its Hilbert series by $1-u^e$. Applying this three times gives

$$H_{\mathbb{J}(\phi)}(u) = H_S(u)(1-u^{d-a})(1-u^{d-b})(1-u^{d-c}) = \frac{(1-u^{d-a})(1-u^{d-b})(1-u^{d-c})}{(1-u^a)(1-u^b)(1-u^c)}.$$

Multiplying both numerator and denominator by $(-1)^3$ gives exactly (12). □

Set $\mathcal{M}_\phi := \phi^{-1}(0) \subset \mathbb{C}^3$. Taking the limit of the right-hand side of (12) as $u \to 1$ gives the formula

$$\dim \mathbb{J}(\phi) = \mu := \frac{(d-a)(d-b)(d-c)}{abc}, \tag{13}$$

for the *Milnor number* of the isolated singularity (at the origin) of the hypersurface $\mathcal{M}_\phi$.

3.3. **Simple elliptic singularities.** In this subsection, we specialize to isolated elliptic singularities. To explain the classification in proposition 3.11, we first pass from the affine hypersurface to a projective curve.

Indeed, there is a general procedure for this. Let

$$\mathbb{P}^{a,b,c} := (\mathbb{C}^3 \setminus \{0\})/\mathbb{C}^\times$$

denote the weighted projective plane corresponding to the $\mathbb{C}^\times$-action with weights (a,b,c), where $\gcd(a,b,c) = 1$. Restricting the projection $\mathbb{C}^3 \setminus \{0\} \to \mathbb{P}^{a,b,c}$ to the punctured hypersurface, one obtains a map

$$\mathcal{M}_\phi \setminus \{0\} \longrightarrow \mathbb{P}(\mathcal{M}_\phi) \subset \mathbb{P}^{a,b,c}.$$

This way $\mathcal{M}_\phi \setminus \{0\}$ becomes a principal $\mathbb{C}^\times$-bundle over $\mathbb{P}(\mathcal{M}_\phi)$, a projective curve. The type of the hypersurface $\mathcal{M}_\phi$ is closely related to the integer

$$\varpi := d - a - b - c. \tag{14}$$

The case $\varpi = 0$ corresponds to simple elliptic singularities $\widetilde{E}_6$, $\widetilde{E}_7$, $\widetilde{E}_8$ (for a reducible curve, all components must be rational). The following statement is recalled in [EG10, Proposition 2.3.2].

Proposition 3.11. *Let the variables x, y, z have degrees $0 < a \le b \le c$, such that $\gcd(a,b,c) = 1$.*

(i) Let $\phi \in \mathbb{C}[x,y,z]$ be an irreducible homogeneous polynomial of degree $\deg\phi \le a+b+c$. Then, the projective curve $\phi(x,y,z) = 0$ is either rational or elliptic.

(ii) Let $d \le a+b+c$ be such that, for a general homogeneous polynomial ϕ of degree d, the projective curve $\phi(x,y,z) = 0$ is elliptic. Then, $d = a+b+c$, and we have:

- One of the following holds:

		a	b	c	d	$p := \frac{d}{a}$	$q := \frac{d}{b}$	$r := \frac{d}{c}$	μ
1.	E_6 *case:*	1	1	1	3	3	3	3	8
2.	E_7 *case:*	1	1	2	4	4	4	2	9
3.	E_8 *case:*	1	2	3	6	6	3	2	10

(15)

moreover, the integers $(p-1, q-1, r-1)$ give the lengths of the three legs of the corresponding extended Dynkin diagram of type $\widetilde{E}_6$, $\widetilde{E}_7$, or $\widetilde{E}_8$.

- *The homogeneous equation of the corresponding elliptic curve can be brought to the canonical form*

$$\phi^\tau(x,y,z) = \frac{x^p}{p} + \frac{y^q}{q} + \frac{z^r}{r} + \tau xyz = 0, \qquad \text{where } \tau \in \mathbb{C}^\times. \tag{16}$$

In the setting of proposition 3.11(ii), we have

$$\frac{1}{p} + \frac{1}{q} + \frac{1}{r} = \frac{a}{d} + \frac{b}{d} + \frac{c}{d} = \frac{a+b+c}{d} = 1.$$

For the Milnor number, this gives

$$\mu = \dim \mathbb{J}(\phi^\tau) = \frac{(a+b)(a+c)(b+c)}{abc} = (p-1)(q-1)(r-1).$$

Since $pq + pr + qr = pqr$, the last expression equals $p + q + r - 1$, in agreement with theorem 3.8.

4. Poisson cohomology

In this section, we study the Poisson structure on a hypersurface in $\mathbb{C}^3$ with an isolated singularity. We start with the Jacobian Poisson bracket on $\mathbb{C}[x,y,z]$. For a polynomial $\phi \in \mathbb{C}[x,y,z]$, it is defined on the coordinate functions by

$$\{x,y\} = \frac{\partial \phi}{\partial z}, \quad \{y,z\} = \frac{\partial \phi}{\partial x}, \quad \{z,x\} = \frac{\partial \phi}{\partial y}.$$

In view of section 2, this Poisson bracket is determined by a bivector $\pi \in \mathfrak{X}^2(\mathbb{C}^3)$ such that $i_\pi \mathrm{vol} = d\phi$, where vol is the nowhere vanishing volume form $dx \wedge dy \wedge dz$. Equivalently, the bracket is determined by the equation

$$\{f,g\} \cdot \mathrm{vol} = d\phi \wedge df \wedge dg, \qquad \forall f, g \in \mathbb{C}[x,y,z].$$

We denote this Poisson algebra on $\mathbb{C}[x,y,z]$ by $\mathscr{A}_\phi$. By the characterization above one sees that ϕ is in the center of $\mathscr{A}_\phi$ (i.e. $\{\phi, f\} = 0$ for all $f \in \mathscr{A}_\phi$). (This can also be checked by direct calculation.) Hence we can consider the quotient Poisson algebra $\mathscr{B}_\phi := \mathscr{A}_\phi/(\phi)$.

The main theorem we want to explain in this section is the following Poisson enhancement of theorem 3.8.

Theorem 4.1. *The family of Poisson algebras* $\{\mathscr{B}_\phi, \phi = c \cdot \phi^{\tau,\nu}_{P,Q,R}, c \in \mathbb{C}^\times\}$ *provides a semiuniversal Poisson deformation of* $\mathscr{B}_{\phi^\tau}$.

As explained in section 1.1, the relevant infinitesimal deformations are governed by Poisson cohomology. We compute the cohomology of B_ϕ using Pichereau's calculations for $\mathscr{A}_\phi$, the Poisson algebra of the ambient space $\operatorname{Spec} \mathscr{A}_\phi = \mathbb{C}^3$; see [Pic06]. We use the hypotheses of those calculations, in particular weighted homogeneity and an isolated singularity, when applying them below. We then describe the Kodaira–Spencer map and prove theorem 4.1.

For a singular algebra, the definition of Poisson cohomology uses the cotangent complex. We do not develop that construction here; see [EG10, §5.1]. If $\operatorname{Spec} A$ is smooth, Poisson cohomology is computed by the cohomological Lichnerowicz–Koszul–Brylinski (LKB) complex:

$$PH^\bullet(A) = H^\bullet(\mathfrak{X}^\bullet(A), L_\pi).$$

For a singular hypersurface, we instead use a differential graded resolution. The differential induced by that resolution must be included in the total complex.

4.1. **Poisson cohomology of a hypersurface.** In view of Pichereau's results concerning the Poisson cohomology of a total space A, a smooth finitely generated Poisson $\mathbb{C}$-algebra, we want to relate Poisson cohomology of a hypersurface with that of the total space in this section. More explicity, we want to know about the Poisson cohomology of an algebra of the form $B_\phi := A/(\phi)$, where ϕ is a regular function in the Poisson center.

We first record the anticommutation relation that we will need below, fitting these differentials into the two differentials of a bicomplex.

Lemma 4.2. *For a central function ϕ, the operators L_π and $i_{d\phi}$ anticommute:*

$$L_\pi i_{d\phi} + i_{d\phi} L_\pi = 0.$$

Proof. By corollary 2.7, $L_\pi i_{d\phi} + i_{d\phi} L_\pi = L_{i_{d\phi}\pi}$. Since ϕ is central, (9) gives $i_{d\phi}\pi = [\pi, \phi]_{\mathrm{S}} = X_\phi = 0$. The identity follows. □

We can now state the main result in this subsection.

Proposition 4.3. *Let $\phi \in Z(\mathbb{C}[\mathcal{M}])$ be a central regular function on $\mathcal{M}$. Assume that ϕ has only isolated singularities and that there exists a vector field $\mathrm{eu} \in \mathfrak{X}^1(\mathcal{M})$ such that*

$$\mathrm{eu}(\phi) = c\phi, \qquad c \neq 0.$$

Then, for the Poisson cohomology of the algebra $B_\phi := \mathbb{C}[\mathcal{M}]/(\phi)$, there is a convergent first-quadrant spectral sequence

$$E_1^{p,q} \Longrightarrow \mathrm{gr}_F^p \, PH^{p+q}(B_\phi),$$

whose E_1-page has the following form; see [EG10, Proposition 5.3.4]:

$$\begin{array}{ccccccccc}
\vdots & & \vdots & & \vdots & & \vdots & & \vdots \\
\uparrow{\scriptstyle L_\pi} & & \uparrow{\scriptstyle L_\pi} & & \uparrow{\scriptstyle L_\pi} & & \uparrow{\scriptstyle L_\pi} & & \uparrow{\scriptstyle L_\pi} \\
\mathfrak{X}^4(B_\phi) & \xrightarrow{0} & 0 & \xrightarrow{0} & 0 & \xrightarrow{0} & \mathbb{J}(\phi)\mathrm{eu} & \xrightarrow{0} & \mathbb{J}(\phi) \\
\uparrow{\scriptstyle L_\pi} & & \uparrow{\scriptstyle L_\pi} & & \uparrow{\scriptstyle L_\pi} & & \uparrow{\scriptstyle L_\pi} & & \\
\mathfrak{X}^3(B_\phi) & \xrightarrow{0} & 0 & \xrightarrow{0} & \mathbb{J}(\phi)\mathrm{eu} & \xrightarrow{0} & \mathbb{J}(\phi) & & \\
\uparrow{\scriptstyle L_\pi} & & \uparrow{\scriptstyle L_\pi} & & \uparrow{\scriptstyle L_\pi} & & & & \\
\mathfrak{X}^2(B_\phi) & \xrightarrow{0} & \mathbb{J}(\phi)\mathrm{eu} & \xrightarrow{0} & \mathbb{J}(\phi) & & & & \\
\uparrow{\scriptstyle L_\pi} & & \uparrow{\scriptstyle L_\pi} & & & & & & \\
\mathfrak{X}^1(B_\phi) & \xrightarrow{0} & \mathbb{J}(\phi) & & & & & & \\
\uparrow{\scriptstyle L_\pi} & & & & & & & & \\
\mathfrak{X}^0(B_\phi) & & & & & & & &
\end{array} \tag{17}$$

The horizontal maps shown here are zero because horizontal cohomology has already been taken. The vertical maps are induced by L_π. The leftmost column is the LKB complex of the Poisson algebra B_ϕ, and $\mathbb{J}(\phi)$ denotes the Jacobi ring of ϕ, as in section 3.2.

Proof. The idea of the proof is to provide a dg-algebra resolution of the non-smooth algebra B_ϕ. Set

$$A := \mathbb{C}[\mathcal{M}], \qquad DA := A \otimes \mathbb{C}[\tau]/(\tau^2),$$

a graded super-commutative algebra in which A is the even part placed in degree 0 and τ is an odd generator of degree -1. Introduce a differential

$$\nabla : DA^\bullet \longrightarrow DA^{\bullet+1}$$

defined as the odd super-derivation satisfying

$$\nabla(A) = 0, \qquad \nabla(\tau) = \phi.$$

Equivalently, $\nabla = \phi\frac{\partial}{\partial\tau}$. As a complex this is the two-term complex $A \xrightarrow{\phi} A$. The differential is multiplication by ϕ, so $H^0(DA) = A/(\phi) = B_\phi$ and $H^i(DA) = 0$ for $i \neq 0$.

We can also extend the Poisson bracket to DA by τ-linearity, making DA a Poisson dg algebra quasi-isomorphic to B_ϕ. Hence

$$PH^\bullet(B_\phi) \cong PH^\bullet(DA).$$

The polyvector fields on DA are generated by those on $\mathcal{M}$ and the derivation with respect to the odd variable τ:

$$\mathfrak{X}^\bullet(DA) \cong \mathfrak{X}^\bullet(\mathcal{M}) \otimes_{\mathbb{C}} \mathbb{C}\left[\tau, \frac{\partial}{\partial\tau}\right]/(\tau^2).$$

For the LKB complex on a dg algebra, the differential is $\widetilde{\nabla} + L_\pi$, where $\widetilde{\nabla}$ is induced by ∇. By definition,

$$\widetilde{\nabla}(P) = [\nabla, P].$$

Here the bracket is the graded extension of the Schouten bracket. For a derivation P, it is the graded commutator of ∇ and P. For a multiderivation, it applies ∇ to the output and subtracts its action in each input, with the signs prescribed by the grading. Thus it is the differential on multiderivations induced by the differential on DA. Using the identity

$$[fX, P] = f[X, P] - (i_{df}P) \wedge X$$

with $f = \phi$ and $X = \partial/\partial\tau$, one finds

$$[\nabla, P] = -(i_{d\phi}P) \wedge \frac{\partial}{\partial\tau}.$$

Indeed $[\partial/\partial\tau, P] = 0$ when either $P = \partial/\partial\tau$ or $P \in \mathfrak{X}^\bullet(A)$.

Writing ∂_τ for $\frac{\partial}{\partial\tau}$, we may therefore express the LKB complex as

$$\left(\mathfrak{X}^\bullet(\mathcal{M}) \otimes \frac{\mathbb{C}[\tau, \partial_\tau]}{(\tau^2)},\ \phi\partial_\tau + \tau i_{d\phi} + L_\pi\right).$$

Let

$$T_\phi := T_{\mathcal{M}}\big|_{\phi^{-1}(0)}, \qquad \Lambda_\phi^\bullet := \Gamma\big(\phi^{-1}(0), \Lambda^\bullet T_\phi\big) \cong B_\phi \otimes_A \mathfrak{X}^\bullet(\mathcal{M}).$$

There is a natural surjective algebra morphism

$$\mathrm{pr} : \mathfrak{X}^\bullet(\mathcal{M}) \otimes \frac{\mathbb{C}[\tau, \partial_\tau]}{(\tau^2)} \longrightarrow \Lambda_\phi^\bullet \otimes \mathbb{C}[\partial_\tau],$$

We will explain this morphism in detail below. Let

$$\Lambda_\phi := \Gamma\big(\phi^{-1}(0), \Lambda^\bullet T_\phi\big) \cong B_\phi \otimes_A \mathfrak{X}^\bullet(\mathcal{M}).$$

The morphism above is explicitly given on simple tensors by

$$x \otimes P(\partial_\tau) + \tau\, y \otimes Q(\partial_\tau) \longmapsto [\overline{x}] \otimes P(\partial_\tau),$$

where $\overline{x}$ denotes the residue class of x in $\Lambda_\phi \cong \mathfrak{X}^\bullet(\mathcal{M})/\phi\mathfrak{X}^\bullet(\mathcal{M})$. It is a morphism of dg algebras.

For brevity, denote the source complex of pr by

$$C^\bullet := \mathfrak{X}^\bullet(\mathcal{M}) \otimes \frac{\mathbb{C}[\tau, \partial_\tau]}{(\tau^2)},$$

equipped with the total differential displayed above. We now show that this morphism is a quasi-isomorphism. Its kernel K fits into the short exact sequence

$$0 \longrightarrow K \longrightarrow C^\bullet \xrightarrow{\mathrm{pr}} \Lambda^\bullet_\phi \otimes \mathbb{C}[\partial_\tau] \longrightarrow 0.$$

Thus it is enough to prove that K is acyclic.

Write $K = K^0 \oplus K^1$, corresponding respectively to terms without and with a factor τ. Decompose the total differential as $D = d_0 + d_1$, where

$$d_0 = \phi\partial_\tau : K^1 \longrightarrow K^0, \qquad d_1 = \tau i_{d\phi} + L_\pi.$$

By the preceding lemma, $d_1^2 = 0$, and

$$d_1 d_0 + d_0 d_1 = 0.$$

To make surjectivity explicit, put

$$V := \mathfrak{X}^\bullet(\mathcal{M}) \otimes \mathbb{C}[\partial_\tau].$$

The definition of pr gives $K^0 = \phi V$ and $K^1 = \tau V$: a term without τ lies in the kernel exactly when all its coefficients are divisible by ϕ, and every term with τ is killed by pr. Thus $d_0(\tau v) = \phi v$. Every element $\phi v \in K^0$ has the preimage $\tau v \in K^1$, which proves surjectivity. Multiplication by ϕ is injective, so $d_0 \colon K^1 \to K^0$ is an isomorphism.

Suppose $Dz = 0$ and write $z = z_0 + z_1$, where $z_i \in K^i$. Then

$$d_1 z_1 = 0, \qquad d_1 z_0 + d_0 z_1 = 0.$$

Using the inverse of d_0 and the anticommutation relation, one constructs $w = d_0^{-1}(z_0)$ with $Dw = z$. Hence every cocycle in K is a coboundary, and pr is a quasi-isomorphism.

Consequently the Poisson cohomology of B_ϕ may be computed as the hypercohomology of the dg algebra represented by the following mixed complex:

$$\begin{array}{ccccc}
\cdots & & \cdots & & \cdots \\
\uparrow{\scriptstyle L_\pi} & & \uparrow{\scriptstyle L_\pi} & & \uparrow{\scriptstyle L_\pi} \\
\Lambda^2_\phi & \xrightarrow{i_{d\phi}} & \Lambda^1_\phi & \xrightarrow{i_{d\phi}} & \Lambda^0_\phi \\
\uparrow{\scriptstyle L_\pi} & & \uparrow{\scriptstyle L_\pi} & & \\
\Lambda^1_\phi & \xrightarrow{i_{d\phi}} & \Lambda^0_\phi & & \\
\uparrow{\scriptstyle L_\pi} & & & & \\
\Lambda^0_\phi & & & &
\end{array}$$

We regard the mixed complex displayed above as a first-quadrant bicomplex K, whose horizontal and vertical differentials are $i_{d\phi}$ and L_π, respectively. The associated spectral sequence satisfies

$$E_1^{p,q} = H^{p,q}(K, i_{d\phi}),$$

and its first differential d_1 is induced by L_π. This is well defined because L_π anticommutes with $i_{d\phi}$.

For each $q \geq 0$, let

$$\Lambda_\phi^{(q)} \colon \Lambda_\phi^q \xrightarrow{i_{d\phi}} \Lambda_\phi^{q-1} \xrightarrow{i_{d\phi}} \cdots \xrightarrow{i_{d\phi}} \Lambda_\phi^1 \xrightarrow{i_{d\phi}} \Lambda_\phi^0$$

denote the complex appearing in the q-th row, where Λ_ϕ^j is placed in degree j. We compute its cohomology in the following claim.

Claim. For every $q \geq 0$, one has

$$H^q\big(\Lambda_\phi^{(q)}, i_{d\phi}\big) \cong \mathfrak{X}^q(B_\phi).$$

Moreover,

$$H^j\big(\Lambda_\phi^{(q)}, i_{d\phi}\big) = 0 \qquad \text{for } j \notin \{0, 1, q\},$$

and

$$H^0\big(\Lambda_\phi^{(q)}, i_{d\phi}\big) \cong \mathbb{J}(\phi).$$

If $q > 1$, then also

$$H^1\big(\Lambda_\phi^{(q)}, i_{d\phi}\big) \cong \mathbb{J}(\phi)\mathrm{eu}.$$

Proof of the claim. Put

$$A = \mathbb{C}[\mathcal{M}], \qquad B_\phi = A/(\phi), \qquad \mathcal{M}_\phi = \phi^{-1}(0).$$

We first prove the assertion concerning the leftmost cohomology. The module of Kähler differentials of B_ϕ has the presentation

$$B_\phi \cdot d\phi \longrightarrow \Omega_A^1 \otimes_A B_\phi \longrightarrow \Omega_{B_\phi}^1 \longrightarrow 0.$$

A q-polyderivation of B_ϕ is equivalently a B_ϕ-linear alternating map

$$\bigwedge^q \Omega_{B_\phi}^1 \longrightarrow B_\phi.$$

Pulling such a map back to $\bigwedge^q(\Omega_A^1 \otimes_A B_\phi)$ gives a section $s \in \Lambda_\phi^q$ which vanishes whenever one of its arguments is $d\phi$. This condition is precisely

$$i_{d\phi} s = 0.$$

Conversely, every section $s \in \Lambda_\phi^q$ satisfying $i_{d\phi} s = 0$ descends through the above presentation to a q-polyderivation of B_ϕ. Therefore

$$\mathfrak{X}^q(B_\phi) \cong \ker\left(i_{d\phi} \colon \Lambda_\phi^q \longrightarrow \Lambda_\phi^{q-1}\right).$$

Since Λ_ϕ^q is the leftmost term of $\Lambda_\phi^{(q)}$, there is no incoming differential, and hence

$$H^q\big(\Lambda_\phi^{(q)}, i_{d\phi}\big) \cong \mathfrak{X}^q(B_\phi).$$

It remains to compute the other cohomology groups. Consider the truncated Koszul complex

$$\mathfrak{X}^{(q)} \colon \mathfrak{X}^q(\mathcal{M}) \xrightarrow{i_{d\phi}} \mathfrak{X}^{q-1}(\mathcal{M}) \xrightarrow{i_{d\phi}} \cdots \xrightarrow{i_{d\phi}} \mathfrak{X}^0(\mathcal{M}).$$

Locally, this is the Koszul complex associated with the partial derivatives of ϕ. Since ϕ has only isolated singularities, these partial derivatives form a regular sequence. Consequently, the full Koszul complex is exact in positive degrees, and its degree-zero cohomology is

$$H^0\big(\mathfrak{X}^{(q)}, i_{d\phi}\big) = \frac{A}{(\partial\phi)} = \mathbb{J}(\phi).$$

After truncating at degree q, exactness can fail only at the two ends of the complex. Hence $\mathfrak{X}^{(q)}$ is exact except possibly at its leftmost and rightmost terms.

Since

$$\Lambda^j_\phi \cong \mathfrak{X}^j(\mathcal{M})/\phi\mathfrak{X}^j(\mathcal{M}),$$

multiplication by ϕ gives a short exact sequence of complexes

$$0 \longrightarrow \mathfrak{X}^{(q)} \xrightarrow{\cdot\phi} \mathfrak{X}^{(q)} \longrightarrow \Lambda^{(q)}_\phi \longrightarrow 0.$$

The corresponding long exact sequence immediately gives

$$H^j\big(\Lambda^{(q)}_\phi, i_{d\phi}\big) = 0 \qquad \text{for } j \notin \{0,1,q\}.$$

For $q > 1$, its final part is

$$0 \longrightarrow H^1\big(\Lambda^{(q)}_\phi, i_{d\phi}\big) \longrightarrow \mathbb{J}(\phi) \xrightarrow{\cdot\phi} \mathbb{J}(\phi) \longrightarrow H^0\big(\Lambda^{(q)}_\phi, i_{d\phi}\big) \longrightarrow 0.$$

Indeed, the two copies of $\mathbb{J}(\phi)$ here are the degree-zero cohomology groups of the two Koszul complexes in the short exact sequence.

By assumption,

$$i_{d\phi}\mathrm{eu} = \mathrm{eu}(\phi) = c\phi, \qquad c \neq 0.$$

Thus

$$\phi = c^{-1} i_{d\phi}\mathrm{eu}$$

belongs to the Jacobi ideal, so its class in $\mathbb{J}(\phi)$ vanishes. Hence multiplication by ϕ on $\mathbb{J}(\phi)$ is the zero map. Exactness therefore yields

$$H^0\big(\Lambda^{(q)}_\phi, i_{d\phi}\big) \cong \mathbb{J}(\phi), \qquad H^1\big(\Lambda^{(q)}_\phi, i_{d\phi}\big) \cong \mathbb{J}(\phi).$$

More precisely, the class of $c^{-1}\mathrm{eu}$ is mapped to $1 \in \mathbb{J}(\phi)$ by the connecting morphism. Up to the nonzero scalar c, this identifies the latter group with

$$\mathbb{J}(\phi)\mathrm{eu}.$$

This proves the claim.

Applying the claim to every row of K gives exactly the E_1-page displayed in (17). The differential d_1 is induced by the remaining differential L_π. Since K is a first-quadrant bicomplex, its standard filtration gives a convergent spectral sequence

$$E_1^{p,q} \Longrightarrow \mathrm{gr}^p_F\, PH^{p+q}(B_\phi),$$

as required. □

4.2. **Pichereau's results.** We now specialize to the quasi-homogeneous case. Thus, let

$$0 < a \leq b \leq c$$

be positive integers such that $\gcd(a,b,c) = 1$. Let eu denote the corresponding Euler vector field on $\mathcal{M} = \mathbb{C}^3$, and let Υ be the standard three-vector $\frac{\partial}{\partial x} \wedge \frac{\partial}{\partial y} \wedge \frac{\partial}{\partial z}$ which is dual to the volume form $dx \wedge dy \wedge dz$.

Given a homogeneous polynomial $\phi \in \mathbb{C}[\mathcal{M}] = \mathbb{C}[x,y,z]$, we put $\pi := i_{d\phi}\Upsilon$, and let $\mathscr{A}_\phi$ denote the corresponding Poisson algebra.

Pichereau computed the Poisson cohomology groups of $\mathscr{A}_\phi$ explicitly [Pic06, Propositions 4.2, 4.5, 4.7, and 4.8]. To state the result, set $\mu := \dim_{\mathbb{C}} \mathbb{J}(\phi)$, the Milnor number, and choose homogeneous elements

$$f_0 = 1, f_1, \ldots, f_{\mu-1} \in \mathbb{C}[x,y,z]$$

whose residue classes modulo the Jacobi ideal form a $\mathbb{C}$-basis of the Jacobi ring $\mathbb{J}(\phi)$. We regard

$$i_{df_k}\Upsilon \in \mathfrak{X}^2(\mathbb{C}^3), \qquad k = 1, \ldots, \mu - 1,$$

as elements of the LKB complex of $\mathscr{A}_\phi$.

Proposition 4.4 (Pichereau)**.** *Let ϕ be a homogeneous polynomial with an isolated singularity and suppose that*

$$\deg \phi = a + b + c.$$

Then the Poisson cohomology of $\mathscr{A}_\phi$ vanishes in degrees greater than or equal to 4. Moreover, the following statements hold.

(i) One has

$$PH^0(\mathscr{A}_\phi) = \mathbb{C}[\phi].$$

Furthermore,

$$PH^1(\mathscr{A}_\phi) \cong \mathbb{C}[\phi]\,[\mathrm{eu}].$$

Here multiplication by a central element induces the $\mathbb{C}[\phi]$-module structure on cohomology. The displayed isomorphism sends a polynomial $h(\phi)$ to the class $[h(\phi)\mathrm{eu}]$; thus the cohomology class [eu] *is a free generator.*

(ii) The group $PH^3(\mathscr{A}_\phi)$ is a free $\mathbb{C}[\phi]$-module of rank μ, with basis represented by

$$\Upsilon,\ f_1\Upsilon,\ \ldots,\ f_{\mu-1}\Upsilon.$$

Similarly, $PH^2(\mathscr{A}_\phi)$ is a free $\mathbb{C}[\phi]$-module with basis represented by

$$i_{df_1}\Upsilon,\ \ldots,\ i_{df_{\mu-1}}\Upsilon,\ \pi.$$

Pichereau also computed the cohomology groups of the LKB complex of the singular Poisson algebra

$$B_\phi := \mathbb{C}[x, y, z]/(\phi)$$

associated with a quasi-homogeneous polynomial $\phi \in \mathbb{C}[x, y, z]$ of arbitrary positive weighted degree d, with an isolated singularity. More precisely,

$$\mathfrak{X}^p(B_\phi) = 0 \qquad \text{for all } p > 2,$$

and the cohomology of the LKB differential L_π is [Pic06, Propositions 5.3, 5.5, and 5.6]

$$\begin{aligned} H^0\big(\mathfrak{X}^\bullet(B_\phi), L_\pi\big) &= \mathbb{C}, \\ H^1\big(\mathfrak{X}^\bullet(B_\phi), L_\pi\big) &= \mathbb{J}^{(\varpi)}(\phi)\,\mathrm{eu}, \\ H^2\big(\mathfrak{X}^\bullet(B_\phi), L_\pi\big) &\cong \mathbb{J}^{(\varpi)}(\phi)\,\pi. \end{aligned} \tag{18}$$

Here

$$\varpi := d - a - b - c,$$

and $\mathbb{J}^{(\varpi)}(\phi)$ denotes the homogeneous component of weight ϖ in the Jacobi ring. Equivalently, with the notation used above,

$$\mathbb{J}^{(\varpi)}(\phi) = \operatorname{span}_{\mathbb{C}} \left\{ \overline{f_j} \;\middle|\; 0 \le j \le \mu - 1,\ \deg f_j = \varpi \right\}.$$

4.3. **Poisson cohomology of B_ϕ and proof of theorem 4.1.** With the results of Pichereau and the preceding proposition at hand, we can calculate the Poisson cohomology of B_ϕ. We refer to [EG10, Theorem 2.4.1(ii) and §5.5] for the proof, which uses the preceding computations.

Theorem 4.5. *Let $\phi \in \mathbb{C}[x,y,z]$ be weighted homogeneous of degree d, with an isolated singularity at the origin. As vector spaces, the Poisson cohomology groups of B_ϕ are*

$$\begin{aligned} PH^0(B_\phi) &= \mathbb{C}, \\ PH^1(B_\phi) &= \mathbb{J}^{(\varpi)}(\phi), \\ PH^2(B_\phi) &= \mathbb{J}^{(\varpi)}(\phi) \oplus \mathbb{J}(\phi), \\ PH^k(B_\phi) &= \mathbb{J}(\phi) \qquad \text{for } k \geq 3, \end{aligned}$$

where

$$\varpi := d - (a+b+c),$$

and the Jacobi ring $\mathbb{J}(\phi)$ is as in section 3.2.

We now use the theorem to prove theorem 4.1.

Proof of theorem 4.1. Let

$$S = \mathbb{C}^2 \times S_p \times S_q \times S_r \times \mathbb{C}^\times,$$

where the parameters τ, ν form coordinates in the first factor $\mathbb{C}^2$, the affine linear spaces S_p, S_q, S_r correspond to the coefficients α_i of P, β_i of Q, and γ_i of R, and the parameter c is a coordinate on the last factor $\mathbb{C}^\times$. This space parametrizes the family

$$\left\{ B_\phi \;\middle|\; \phi = c\,\phi^{\tau,\nu}_{P,Q,R},\ c \in \mathbb{C}^\times \right\}.$$

Fix $\tau_0 \in \mathbb{C}^\times$ such that ϕ^{τ_0} has an isolated singularity. Let $o \in S$ be the point where $\tau = \tau_0$, $c = 1$, $\nu = 0$, and all coefficients $\alpha_i, \beta_i, \gamma_i$ vanish. The fiber at o is $B_{\phi^{\tau_0}}$. We work locally near o and abbreviate ϕ^{τ_0} to ϕ^τ below.

By definition, we need to show that the Kodaira–Spencer map

$$T_o S \longrightarrow PH^2(B_{\phi^\tau})$$

is an isomorphism. By the preceding theorem,

$$PH^2(B_{\phi^\tau}) \cong \mathbb{J}(\phi^\tau) \oplus \mathbb{C},$$

where the second summand is spanned by the cohomology class of the Poisson bivector π. It corresponds to infinitesimally rescaling the bracket. Moreover,

$$\dim T_o S = (p-1) + (q-1) + (r-1) + 3 = \mu + 1 = \dim PH^2(B_{\phi^\tau}),$$

so it remains to prove that the Kodaira–Spencer map is injective.

Explicitly, a tangent vector to S is represented by a polynomial

$$\dot{F} = \dot{c}\,\phi^\tau + \dot{\tau}\,xyz + \dot{P} + \dot{Q} + \dot{R} + \dot{\nu}.$$

Equivalently, introducing a formal variable ε with $\varepsilon^2 = 0$, the tangent polynomial $\dot{F}$ is the coefficient of ε in $F_\varepsilon = F_0 + \varepsilon \dot{F}$, where

$$\begin{aligned} F_0 &= \phi^\tau, \\ F_\varepsilon &= (1 + \varepsilon\dot{c}) \left(\phi^\tau + \varepsilon(\dot{\tau}\,xyz + \dot{P} + \dot{Q} + \dot{R} + \dot{\nu}) \right). \end{aligned}$$

Writing $\dot{F} = \dot{c}\,\phi^\tau + \dot{F}'$, the Kodaira–Spencer map, with the splitting determined by these deformation directions, sends this tangent vector to

$$\left([\dot{F}'], \dot{c} \right) \in \mathbb{J}(\phi^\tau) \oplus \mathbb{C}.$$

It remains to show that the residue classes of

$$1,\ x,\dots,x^{p-1},\ y,\dots,y^{q-1},\ z,\dots,z^{r-1},\ xyz$$

are linearly independent in $\mathbb{J}(\phi^\tau)$. There are exactly μ of them, and the Jacobi ring has dimension μ. We will show that they span. This also supplies the details of the injectivity argument.

In the Jacobi ring, the three defining relations are

$$x^{p-1} = -\tau yz, \qquad y^{q-1} = -\tau xz, \qquad z^{r-1} = -\tau xy.$$

The isolated-singularity assumption gives $1+\tau^3 \neq 0$ in type E_6, $1-\tau^4 \neq 0$ in type E_7, and $1-\tau^6 \neq 0$ in type E_8. Indeed, if the corresponding factor vanishes, these three partial derivatives vanish simultaneously at a nonzero point; weighted homogeneity then gives a nonisolated critical locus.

Type E_6: $(p,q,r) = (3,3,3)$. Here $x^2 = -\tau yz$, $y^2 = -\tau xz$, and $z^2 = -\tau xy$. For example,

$$x^2y = -\tau y^2 z = \tau^2 xz^2 = -\tau^3 x^2 y,$$

so $x^2y = 0$. The same calculation applies to every monomial with one variable squared and a different variable to the first power. Also $x^3 = y^3 = z^3 = -\tau xyz$. It follows that every monomial of degree at least four vanishes, and the ring is spanned by

$$1, x, y, z, xy, xz, yz, xyz.$$

Since $\tau \neq 0$, the three mixed quadratics can be replaced by x^2, y^2, z^2. This gives the required spanning set.

Type E_7: $(p,q,r) = (4,4,2)$. Eliminating $z = -\tau xy$ gives

$$x^3 = \tau^2 xy^2, \qquad y^3 = \tau^2 x^2 y.$$

Hence $x^3y = \tau^2 xy^3 = \tau^4 x^3 y$, so $x^3y = xy^3 = 0$. Furthermore, $x^4 = y^4 = \tau^2 x^2 y^2$, and all monomials in x, y of degree at least five vanish. The ring is therefore spanned by

$$1, x, y, x^2, xy, y^2, xy^2, x^2y, x^2y^2.$$

The identities

$$z = -\tau xy, \quad x^3 = \tau^2 xy^2, \quad y^3 = \tau^2 x^2 y, \quad xyz = -\tau x^2 y^2$$

show that this span is also generated by $1, x, x^2, x^3, y, y^2, y^3, z, xyz$, as required.

Type E_8: $(p,q,r) = (6,3,2)$. Again eliminate $z = -\tau xy$. The remaining relations become

$$y^2 = \tau^2 x^2 y, \qquad x^5 = \tau^4 x^3 y.$$

Multiplying the second relation by y gives $x^5y = \tau^6 x^5 y$, so $x^5y = 0$ and then $x^7 = 0$. The first relation reduces any power of y to a multiple of y; the second reduces x^5 and x^6 to multiples of x^3y and x^4y. Thus the ring is spanned by the ten monomials

$$1, x, x^2, x^3, x^4,\ y, xy, x^2y, x^3y, x^4y.$$

Using

$$z = -\tau xy, \quad y^2 = \tau^2 x^2 y, \quad x^5 = \tau^4 x^3 y, \quad xyz = -\tau^3 x^4 y,$$

we obtain precisely the span of $1, x, x^2, x^3, x^4, x^5, y, y^2, z, xyz$.

In each case, the μ parameter monomials span the μ-dimensional Jacobi ring, so they form a basis. Therefore $[\dot{F}'] = 0$ forces all coefficients of $\dot{F}'$ to vanish. Together with $\dot{c} = 0$, this proves that the Kodaira–Spencer map is injective. Since its source and target have the same dimension, it is an isomorphism.

□

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References

[EG10] Pavel Etingof and Victor Ginzburg. Noncommutative del Pezzo surfaces and Calabi–Yau algebras. *Journal of the European Mathematical Society*, 12(6):1371–1416, 2010. doi:10.4171/JEMS/235.

[Loo84] Eduard J. N. Looijenga. *Isolated Singular Points on Complete Intersections*, volume 77 of *London Mathematical Society Lecture Note Series*. Cambridge University Press, Cambridge, 1984. See Chapter 6, especially Section 6.C; doi:10.1017/CBO9780511662720.

[Man86] Yuri I. Manin. *Cubic Forms: Algebra, Geometry, Arithmetic*, volume 4 of *North-Holland Mathematical Library*. North-Holland, Amsterdam, 2 edition, 1986. ISBN 0-444-87823-8.

[Pic06] Anne Pichereau. Poisson (co)homology and isolated singularities. *Journal of Algebra*, 299(2):747–777, 2006. doi:10.1016/j.jalgebra.2005.10.029.

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